Caden Franklin

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Education

Rochester Institute of Technology Rochester, NY Software Engineering B.S | Graduating Class of 2026

Relevant Coursework: Web Engineering | Engineering of Software Subsystems | Software Design For Computing Systems | Software Testing

Skills

Programming Languages: Python, Java, SQL, C, C++, HTML, JavaScript
Language-Specific Framework: Angular.js, React.js, PyTorch, JUnit, Flask-vite, Node.js
Operating Systems: Microsoft Windows, Ubuntu Linux
Development Tools: Git, PostgreSQL, Visual Studio, Visual Studio Code

Professional Experience

Integer Technologies LCC | *Software Engineering* | Columbia, South Carolina | *January 2025 - August 2025*

- Improved route optimization for Navy ships by analyzing historical and real-time weather data, contributing to a predictive modeling system that enhanced navigational efficiency
- Developed predictive simulations for energy management systems, enabling proactive adjustments that increased system reliability and reduced fuel consumption

Rochester Institute of Technology Senior Project | *Software Engineering* | Rochester, New York | *January 2026 - August 2026*

- Co-developed the Cognitive Work Analysis Software Tool as part of a six-person team, engineering an asynchronous Flask-based pipeline to automate the extraction of complex system data for Cognitive Work Analysis (CWA).
- Collaborated to architect an NLP-driven framework that converts technical documentation into structured CWA artifacts, reducing manual analysis time and enhancing workflow efficiency for domain experts.

Personal Projects

MAHJONG TILE DETECTOR | *Software Engineering* | Rochester, NY
Technologies Used: Python, Pytorch, OpenCV

- Building a computer vision system capable of detecting Mahjong tiles in live video feeds using a PyTorch-based AI model, increasing object recognition accuracy through optimized video preprocessing with OpenCV

MIDI SONG PLAYER | *Software Engineering* | Rochester, NY
Technologies Used: C, STM32CUBEIDE, TeraTerm terminal emulator, STM 32-bit Arm Cortex MCU

- Programmed an embedded system to control music playback with hardware-based feedback, using LEDs to visually reflect states (Play, Pause, Stop), enhancing user interaction via TeraTerm or physical input

SPORTS DORKS| *Software Engineering* | Rochester, NY
Technologies Used: Java, Angular.js

- Led backend development in a 4-person team to launch a full-stack web app for renting sports equipment and facilities, meeting sprint goals using Agile tools like Trello, GitHub, and Slack
- Increased team efficiency by organizing and delegating tasks based on evolving project requirements

THE ONION | *Software Engineering* | Bronx, NY
Technologies Used: Python, PostgreSQL, PyFlask

- Created a full-stack recipe database website with dynamic filtering and user-friendly UI, improving recipe organization and allowing users to quickly fetch recipes.